insulin has profound effects on glucose metabolism in liver cells, stimulating glycogenesis and inhibiting glycogenolysis, the breakdown of glycogen into glucose.

The pathway of glycogenesis includes a series of steps that result in complex glycogen formation in the cytoplasm of the liver and cells of the muscles. The steps of glycogenesis are as follows:

Step 1: Glucose phosphorylation – In the initial phase, glucose is phosphorylated into glucose-6-phosphate, a usual reaction in glycolysis. It is catalyzed by glucokinase (liver) and hexokinase (muscle).

Step 2: Conversion of Glucose-6-Phosphate to Glucose-1-Phosphate – An enzyme Phosphoglucomutase will catalyze the conversion of glucose-6-P is to glucose-1-phosphate.

Step 3: UTP (uridine triphosphate) attaches to Glucose-1-phosphate in the third step. This step focuses on the reaction of glucose-1-phosphate to UTP thereby forming active nucleotide UDP-Glucose (Uridine diphosphate glucose). The one responsible for such reaction is the enzyme UDP Glucose pyrophosphorylase.

Step 4: UDP-Glucose attaches to glycogen primer, a small fragment of already existing glycogen that serves as a primer, in order to stimulate the synthesis of glycogen. The glucose from UDP-Glucose will be accepted by glycogenin (a protein that acts as a primer). The initial glucose unit is attached to the hydroxyl group of tyrosine of glycogenin. The first molecule of glucose is transferred to glycogenin, which will then take up for glucose residues forming a fragment of primer.

Step 5: Glycogen synthase transfers glucose from UDP-Glucose to glycogen (non-reducing end) forming alpha 1,4-linkages. The same enzyme catalyzes the synthesis of the unbranched molecule with alpha-1,4-glycosidic linkages.

Step 6: The formation of glycogen branches – The final step is the formation of glycogen branches caused by the effect of branching enzyme, which transfers a small fragment of about five to eight residues of glucose from the non-reducing end of the glycogen chain to another glucose residue linked by alpha-1,6 bond. This action causes the formation of a new non-reducing end. The final result is the elongation and branching out of the glycogen chain.

The process of glycogenesis utilizes two molecules of ATP. One molecule is needed for glucose phosphorylation and another molecule is needed to convert UDP to UTP.

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